

# Vishal Kumar Sharma

Application Manager, Legal & General · PhD Researcher · Author

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## Profile

Analytical data scientist and strategist with **11 years** of cross-industry experience in analytics, machine learning and predictive modelling across insurance, banking, real estate and manufacturing. Builds production analytics products and interactive R Shiny applications that drive business decisions, backed by doctoral research in **multimodal deepfake detection** and peer-reviewed work in IEEE and Springer venues.

## Experience

**Application Manager** **Apr 2023 – Present**
*Legal & General Insurance* *United Kingdom*

- Lead development of risk-accumulation analytics: Shiny applications for underwriters with interactive mapping (leaflet) and visualisation (plotly, highcharts).
- Built a GIP claims-prediction model (binary classification in Dataiku and R) with full EDA and stakeholder reporting.

**Senior Analyst – Pricing / R Shiny Developer** **Jul 2021 – Apr 2023**
*Equipped Analytical Intelligence*

- Priced portfolios for investment opportunities of up to **£150M**, including NPL and securitisation pricing with risk–return analysis.
- Developed predictive models for loan defaults, collections and prepayments; led dashboard-optimisation initiatives.

**Data Scientist** **Jan 2020 – Jun 2021**
*CBRE*

- Built the Portfolio Transformation Calculator for COVID-19 impact analysis and the ML-powered *CBRE Pathfinder* dashboard (Python–R via Flask APIs on AWS).
- Delivered location-intelligence systems (Google Maps API), automated EDA tools and portfolio dashboards for Fortune 500 clients.

**Junior Data Scientist** **Jul 2015 – Jan 2020**
*Sopra Banking Software*

- Banking platform development for NS&I (UK); churn and subscription-likelihood models with interactive R Shiny campaign analysis.
- Developed house-price prediction and customer-segmentation models.

## Education

**Ph.D., Computer Science and Engineering** **2021 – 2026**
*Amity University, Noida*
Thesis: *Detecting DeepFake Videos Based on Spatio-temporal Features & Audio-Visual Modalities Using Artificial Intelligence*

**Masters in Software Systems (Data Science)** **2017 – 2020**
*BITS Pilani* *CGPA 8.47/10*

**B.Tech., Computer Science & Engineering** **2011 – 2015**
*Inderprastha Engineering College, Ghaziabad* *72.9%*

## Technical Skills

| | |
|---|---|
| **Languages** | R, Python, SQL, JavaScript, Java, Shell scripting |
| **ML & AI** | Machine learning, deep learning, statistical modelling, NLP, computer vision, prompt engineering |
| **Platforms** | R Shiny, Dataiku, Snowflake, AWS, TensorFlow, Tableau, Power BI, Jupyter |
| **Visualisation** | ggplot2, Plotly, Leaflet, Highcharts, D3.js |

## Key Projects

- **Accumulation of Risk Analytics** (insurance) – Shiny application for underwriters to calculate risk accumulation, with interactive leaflet mapping.
- **GIP Claims Prediction** (machine learning) – classification model predicting claim-payment likelihood, built in Dataiku and R.
- **Portfolio Pricing Engine** (financial analytics) – NPL and securitisation pricing for opportunities up to £150M with risk–return analysis.
- **CBRE Pathfinder** (real-estate tech) – ML-powered property-decision dashboard integrating Python and R through Flask APIs on AWS.
- **Banking Churn Prediction** (banking) – end-to-end churn model for NS&I with R Shiny visualisations and automated campaign analysis.

## Book

**Prompt Engineering for Production Systems: Building Reliable, Testable, and Maintainable LLM Applications** (First Edition, 2026). A practitioner's guide to evaluation, regression testing, reliability, observability and security for LLM applications. Available on Amazon.

## Selected Publications

**2026** **Point-based Spatio-temporal Convolution and Dual-channel Attention for Lightweight Multimodal Deepfake Detection.** V. K. Sharma, S. Rawat, Q. Caudron. *Iranian Journal of Science and Technology, Trans. Electrical Engineering*, Springer. doi:10.1007/s40998-026-01247-4

**2026** **Audio-Visual Fusion for Deepfake Video Detection: A Deep Learning Framework.** V. K. Sharma, R. Garg, Q. Caudron. ICAAAIML 2023, *Lecture Notes in Electrical Engineering* vol. 1464, Springer, pp. 119–133. doi:10.1007/978-981-95-0148-9_10

**2025** **Enhancing Deepfake Detection Through Dynamics of Facial Expressions.** V. K. Sharma, S. Rawat. *6th Int. Conf. on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV)*, IEEE. doi:10.1109/ICICV64824.2025.11085942

**2024** **A Systematic Literature Review on Deepfake Detection Techniques.** V. K. Sharma, R. Garg, Q. Caudron. *Multimedia Tools and Applications*, Springer. doi:10.1007/s11042-024-19906-1

**2024** **Stronger Together: Crafting Successful Industry–Academia Partnerships.** Springer book chapter. doi:10.1007/978-981-96-0201-8_51

**2024** **Spatio-Temporal Convolutional Neural Networks for Deepfake Detection: An Empirical Study.** V. K. Sharma, R. Garg, Q. Caudron. *2nd Int. Conf. on Informatics (ICI)*, IEEE. doi:10.1109/ICI60088.2023.10420892

## Recognition

- **Best Shiny App Award** – first prize for the best Shiny application within the organisation.
- **Customer Delight Award** – for consistency and commitment to quality deliveries.